

Module 5: Testing, Quality & Maintenance

Black-Box (ECP/BVA), White-Box (Basis Path & Cyclomatic Complexity), Mutation & 4 Maintenance Types

Module V: Project Management, Quality & Maintenance — Complete 13-Topic Tracker

- Topic 56:** Process & Product Metrics (Defect Density & Productivity)

Topic 58: Software Project Estimation Paradigms & Heuristics

Topic 60: Empirical Estimation Models & Historical Calibration

Topic 62: Function Point Analysis (5 Function Types: EI, EO, EQ, ILF, EIF)

Topic 64: Quality Planning & SQA Checklists

Topic 66: Software Configuration Management (SCM, Baselines & CCB)

Topic 68: Software Re-Engineering (Reverse & Forward Engineering)
- Topic 57:** Software Measurement (Lines of Code & Token Metrics)

Topic 59: Decomposition Techniques (Problem vs. Process Decomposition)

Topic 61: The COCOMO Model (Organic, Semi-Detached, Embedded)

Topic 63: Quality Assurance & Standards (ISO 9001 & CMMI 5 Levels)

Topic 65: Quality Control (QA vs. QC: Prevention vs. Detection)

Topic 67: Software Maintenance (The 4 CAP-P Maintenance Types)

Topic 61: Boehm's COCOMO Model (Constructive Cost Model)

Basic COCOMO Equations (Barry Boehm, 1981):

$$\text{Effort } (E) = a \cdot (\text{KLOC})^b \quad [\text{Person-Months (PM)}]$$
$$\text{Development Time } (T_{\text{dev}}) = c \cdot (E)^d \quad [\text{Months}]$$
$$\text{Average Staffing Size } (SS) = \frac{E}{T_{\text{dev}}} \quad [\text{Persons}], \quad \text{Productivity} = \frac{\text{KLOC}}{E} \quad [\text{KLOC/PM}]$$

Project Mode	Team & Environment Characteristics	Effort (a, b)	Time (c, d)
1. Organic	Small, experienced teams working with flexible requirements in familiar in-house environment.	$E = 2.4(\text{KLOC})^{1.05}$	$T_{\text{dev}} = 2.5(E)^{0.38}$
2. Semi-Detached	Medium teams with mixed experience levels working under intermediate constraints.	$E = 3.0(\text{KLOC})^{1.12}$	$T_{\text{dev}} = 2.5(E)^{0.35}$
3. Embedded	Extremely tight, rigid hardware/software interface constraints (e.g., flight control, medical systems).	$E = 3.6(\text{KLOC})^{1.20}$	$T_{\text{dev}} = 2.5(E)^{0.32}$

Topic 62: Function Point (FP) Analysis (Albrecht, 1979)

Function Point Mathematical Formulations:

$$FP = UFP \times VAF = UFP \times \left[0.65 + 0.01 \times \sum_{i=1}^{14} C_i \right]$$

Where **UFP** is Unadjusted Function Points, and $\sum C_i$ is the sum of 14 General System Characteristics (TDI: 0 to 70).

Function Type	Low Weight	Average Weight	High Weight	Description
1. External Inputs (EI)	3	4	6	User transactions adding/updating internal DB records.
2. External Outputs (EO)	4	5	7	Calculated reports/displays leaving system boundary.
3. External Inquiries (EQ)	3	4	6	Interactive queries retrieving direct data without calculation.
4. Internal Logical Files (ILF)	7	10	15	Major database tables maintained inside system boundary.
5. External Interface Files (EIF)	5	7	10	Data files maintained by another external system.

Topic 63: Capability Maturity Model Integration (CMMI Levels 1 – 5)

CMMI Maturity Level	Process Predictability & Characteristics
Level 1: Initial	Ad-hoc, chaotic, undocumented; success depends on individual heroic developer efforts.
Level 2: Managed	Basic project management established; requirements, costs, and schedules are tracked and repeatable.
Level 3: Defined	Standard engineering processes documented and standardized across entire organization.
Level 4: Quantitatively Managed	Process and product quality quantitatively measured, controlled, and statistically tracked.
Level 5: Optimizing	Continuous process improvement driven by quantitative feedback and innovative technology piloting.

Topic 66 – 68: SCM & The 4 Types of Software Maintenance (CAP-P)

 **Memory Hook: The 4 Maintenance Types (CAP-P)**

- 1. **Corrective Maintenance (20%)**: Reactive fixing of residual bugs discovered in production.
- 2. **Adaptive Maintenance (25%)**: Modifying software to operate in new environments (OS upgrade, cloud migration).
- 3. **Perfective Maintenance (50% — Largest)**: Enhancing existing functionality, performance, and user convenience.
- 4. **Preventive Maintenance (5%)**: Refactoring code to enhance maintainability and prevent future software decay.

M5 Active Recall & Exam Questions

Q1. A 30 KLOC software is developed in Organic mode. Calculate Effort, Development Time, Staff Size, and Productivity using Basic COCOMO. (8 Marks)

Given: Size = 30 KLOC, Mode: Organic ($a = 2.4, b = 1.05, c = 2.5, d = 0.38$).

1. Effort $E = 2.4 \times (30)^{1.05} = 2.4 \times 35.64 \approx \mathbf{85.54}$ Person-Months.
2. Development Time $T_{\text{dev}} = 2.5 \times (85.54)^{0.38} = 2.5 \times 5.43 \approx \mathbf{13.58}$ Months.
3. Average Staff Size $SS = \frac{E}{T_{\text{dev}}} = \frac{85.54}{13.58} \approx \mathbf{6.3}$ Persons.
4. Productivity $= \frac{30}{85.54} \approx \mathbf{0.351}$ KLOC/PM (**351** LOC/PM).